

Description:

1.1 Product Function

DCM dispersion compensator has the function of slope compensation. It can compensate the dispersion slope of the standard singlemode fiber (G.652) in the C band, which can optimize the residual dispersion. It is based on mature and reliable optical fiber technology, which can improve the performance of the optical transmission system. The dispersion of the 1550nm wavelength can reach -10 to -2100ps/nm and can provide products with special requirements for central wavelength and dispersion.



1.2 Product Features

- G.652 fiber C band 100% slope compensation (Standard Value)
- Low polarization mode dispersion
- Wide band dispersion compensation for DWDM system
- Performance index through the GR-2854-CORE Telcordia standard certification
- Reliability through the GR-2854-CORE Telcordia standard certification

2.0 Optical Specification

Parameters	Unit	DCM-20	DCM-30	DCM-40	DCM-50	DCM-60	DCM-70	Notes
Compensated Distance	km	20	30	40	50	60	70	1
Dispersion@1545nm	ps/nm	-340±3%	510±3%	-680±3%	850±3%	-1020±3%	-1190±3%	2
RDS	1/nm	0.0036±20%						3
Max. Insertion Loss (C-Band)	dB	3.5		5.2		6.8	8	4
Typical Insertion Loss (C-Band)	dB	≤2.9	≤3.5	≤4.8	≤5.2	≤6.8	≤7.5	
WDL (C-Band)	dB	≤0.5, type 0.3						
Max. PMD	ps	0.6		0.8		1	1.1	
Typical PMD	ps	≤0.6	≤0.7	≤0.8	≤0.9	≤1.0	≤1.1	
PDL	dB	≤0.1	≤0.1	≤0.1	≤0.1	≤0.1	≤0.1	5
Parameters	Unit	DCM-80	DCM-90	DCM-100	DCM-120	DCM-140	Notes	
Compensated Distance	km	80	90	100	120		1	
Dispersion@1545nm	ps/nm	-1360±3%	-1530±3%	-1700±3%	-2040±3%	-2380±4%	2	
RDS	1/nm	0.0036±20%					3	
Max. Insertion Loss (C-Band)	dB	8	9.5	9.5	11		4	
Typical Insertion Loss (C-Band)	dB	≤8.7	≤9.6	≤10.7	≤14.8	≤16.8		
WDL (C-Band)	dB	≤0.5, type 0.3						
Max. PMD	ps	1.1	1.2	1.2	1.2			
Typical PMD	ps	≤1.2	≤1.3	≤1.4	≤1.5	≤1.6		
PDL	dB	s	≤0.1	≤0.1	≤0.1	≤0.1	5	

Notes:

1. The compensated distance is estimated based on G.652 fiber.
2. This dispersion coefficient is the value of 1545nm wavelength on G.652 fiber.
3. Relative Dispersion Slope is defined as the ratio of dispersion slope to dispersion, RDS=Slope/Dispersion.
4. IL include fiber loss, splice loss and connector – connector interface loss.
5. Polarization dependent loss is the loss deviation induced by two polarization mode.
6. Compensated distance could be customized.
7. Parameter is test at 25 deg.
8. 100% slope compensation of G.652 fiber in C-Band.

2.2 Nonlinear Properties

Parameters	Unit	Min	Max	Notes
SBS Threshold	dBm	4	-	
Nonlinear Coefficient	W ⁻¹	-	1.4x10 ⁻⁹	
Effective Area	μm ²	20	-	

3. Operation/Storage Temperature/Humidity

No	Parameter	Specification	Units	Notes
3.1	Operation Temperature	-5 - +70	°C	
3.2	Operation Humidity	85	%RH	
3.3	Storage Temperature	-40 - +85	°C	
3.4	Storage Humidity	90	%RH	

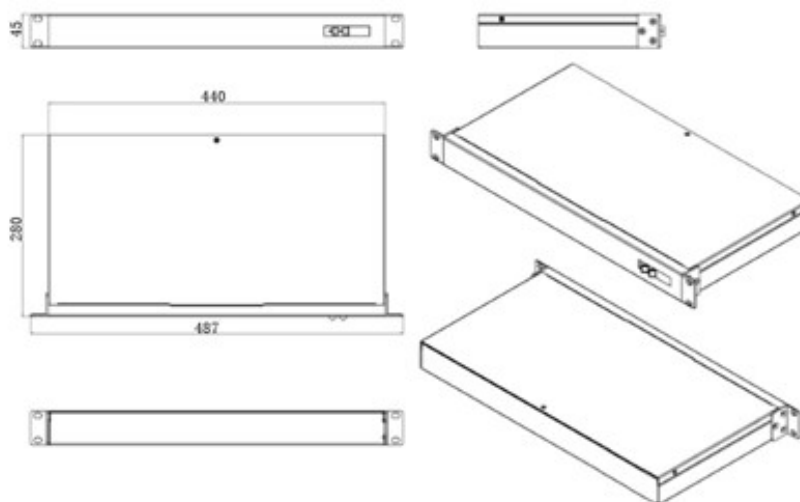
4. Optic Ports Definition

No	Port	Definition of Port	Connector Type	Notes
4.1	1	In	LC/UPC	Or Customized
4.2	2	Out	LC/UPC	Or Customized

5. Pigtail and Connector

No	Parameter	Specification	Unit	Notes
6.1	Pigtail Type (All Ports)	Φ2	mm	Or Customized
6.2	Pigtail Length (All Ports)	0.9±0.1	m	Or Customized
6.3	Connector Type (All Ports)	LC/UPC		Or Customized

Mechanical Dimension



Ordering Information

Part Number	Description
FDCM020F	Fiber based Dispersion Compensation Module, 20 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM030F	Fiber based Dispersion Compensation Module, 30 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM040F	Fiber based Dispersion Compensation Module, 40 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM050F	Fiber based Dispersion Compensation Module, 50 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM060F	Fiber based Dispersion Compensation Module, 60 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM080F	Fiber based Dispersion Compensation Module, 80 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM100F	Fiber based Dispersion Compensation Module, 100 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM120F	Fiber based Dispersion Compensation Module, 120 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"
FDCM140F	Fiber based Dispersion Compensation Module, 140 Km , LC Connectors, C-Band, 1U Rack mount 19"x10"x1.75"

Dual LGX with SC/APC Connectors

Compensation Distance (km)	Dispersion (ps/nm)	Insertion Loss (dB*)	PMD (ps)	PDL (dB)
20	-340±3%	1.9	≤0.6	≤0.1
30	-510±3%	2.8	≤0.7	≤0.1
40	-680±3%	3.2	≤0.8	≤0.1
50	-850±3%	3.5	≤0.9	≤0.1
60	-1020±3%	3.7	≤0.10	≤0.1
70	-1190±3%	4.2	≤0.11	≤0.1
80	-1360±3%	4.6	≤0.12	≤0.1



Physical	
Dimensions (cm.)	12.9H, 15.2D, 5.8W
Dimensions (in.)	5.1H, 6.0D, 2.2W

Environmental	
Storage Temperature	-40° C to 80° C
Operating Temperature	-40° C to 60° C

Ordering Information

Part Number	Description
FDCML020F	Dual LGX with SC/APC Connectors 5.1"x6.0"x2.28 " (HxDxW)
FDCML030F	Dual LGX with SC/APC Connectors 5.1"x6.0"x2.28 " (HxDxW)
FDCML040F	Dual LGX with SC/APC Connectors 5.1"x6.0"x2.28 " (HxDxW)
FDCML050F	Dual LGX with SC/APC Connectors 5.1"x6.0"x2.28 " (HxDxW)
FDCML060F	Dual LGX with SC/APC Connectors 5.1"x6.0"x2.28 " (HxDxW)
FDCML070F	Dual LGX with SC/APC Connectors 5.1"x6.0"x2.28 " (HxDxW)
FDCML080F	Dual LGX with SC/APC Connectors 5.1"x6.0"x2.28 " (HxDxW)